# QSD165D-50 10000W

# Mid-Mounted Motor Product Specification

# 1.Main Parameters

| QSD165D-50 10000W Mid-Mounted Motor | | | |
|---|---|---|---|
| No. | Item | Parameter | Remarks |
| 1 | Motor Type | D165D-50 Mid mounted IPM Permanent Magnet Synchronous DC Motor | PMSM |
| 2 | Rated Power | 10000W | |
| 3 | Peak Power | 20000W | |
| 4 | Rated Voltage | 72V | |
| 5 | Rated Speed | 6200rpm | Motor no-load speed 10000rpm |
| 6 | Maximum Efficiency | ≥94% | |
| 7 | Rated Torque | 15 | |
| 8 | Maximum Torque | ＞73Nm | |
| 9 | Maximum Controller Current Limit | 300A | |
| 10 | Controller Phase Current Peak | 1000A | |
| 11 | Magnet Height | 50H | |
| 12 | Magnet Grade | 38UH | |
| 13 | Number of Pole Pairs | 5 | |
| 14 | Lead Wire Diameter | 30mm² | |
| 15 | Sensor Type | Encoder/ Hall | optional |
| 16 | Temperature Sensor | KTY83-122 | |
| 17 | Waterproof Rating | IP67 | |
| 18 | Motor Weight | 11.26KG | |
| 19 | Duty Cycle | S9 | |

| | | | |
|---|---|---|---|
| 20 | Cooling Method | Natural Cooling | |
| 21 | Maximum Operating Ambient Temperature of Motor | 60℃ | |
| 22 | Maximum Temperature Resistance of Motor Coil | 180℃ | |
| 23 | Temperature Resistance of Motor Enameled Wire | 150℃ | |
| 24 | Insulation Class | Class H | |
| 25 | Motor Slot-Pole | 12S10P | |

## 2.Parameters of the motor's overall dimensions：

Diagram 1：Mid mounted D165D-50 drawing

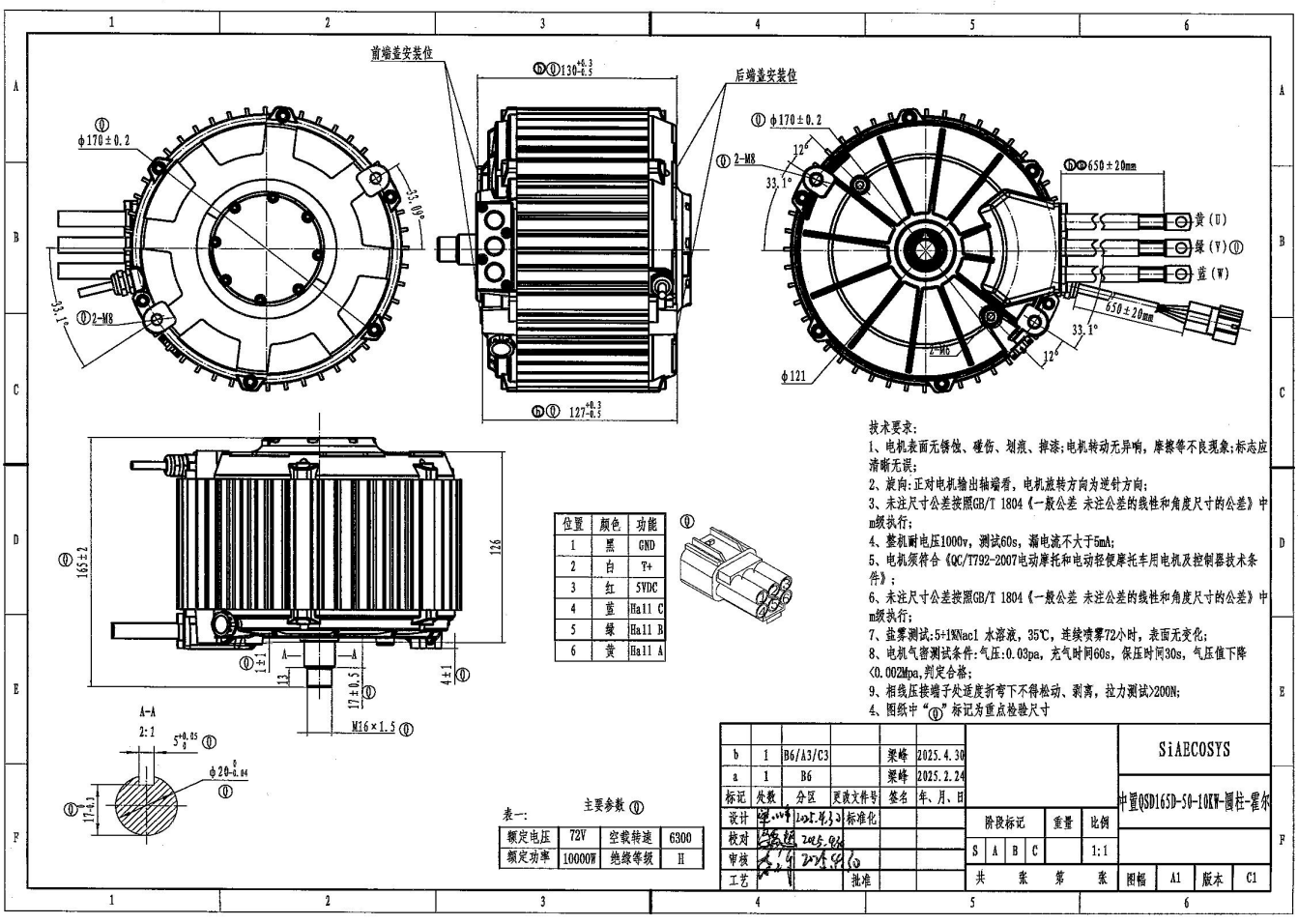



Figure 1

The connection at the motor shaft end in Figure 2 can be customized with sprocket transmission.

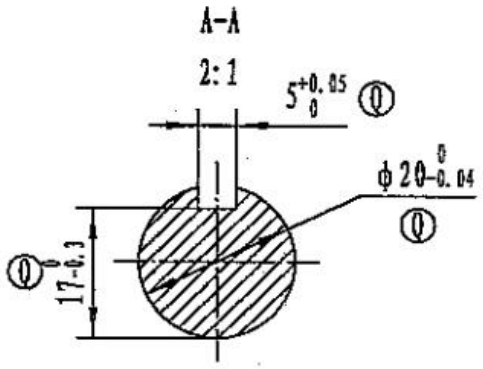
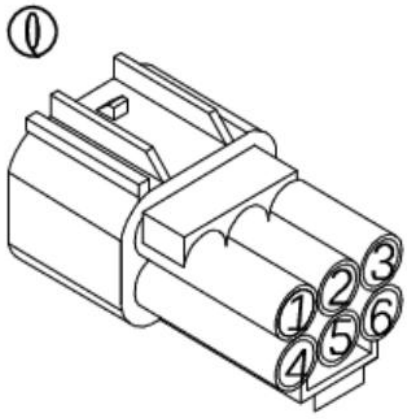



Figure 2

# 3.Motor Wiring Definition

| No. | Color | Definition |
|---|---|---|
| U | Yellow | Phase U |
| V | Green | Phase V |
| W | Blue | Phase W |
| Encoder and Temperature Sensor Output Wire | Black | Encoder and Temperature Sensor |

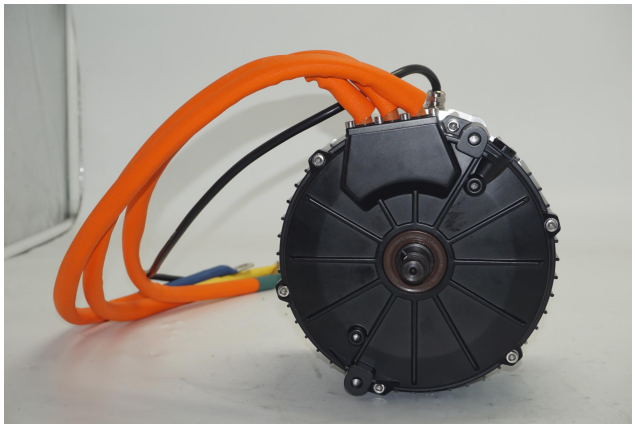
# 4.Terminal model and pin definition

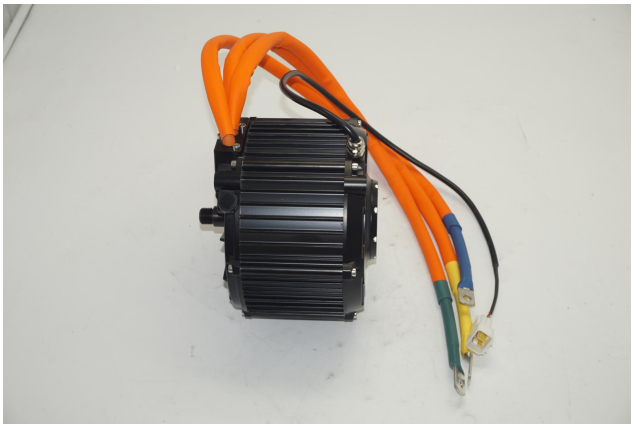


| Position | Color | Function Definition |
|---|---|---|
| 1 | Black | GND |
| 2 | White | T+ |
| 3 | Red | 5VDC |
| 4 | Blue | Hall C |
| 5 | Green | Hall B |
| 6 | Yellow | Hall A |

# 5.Motor Photo

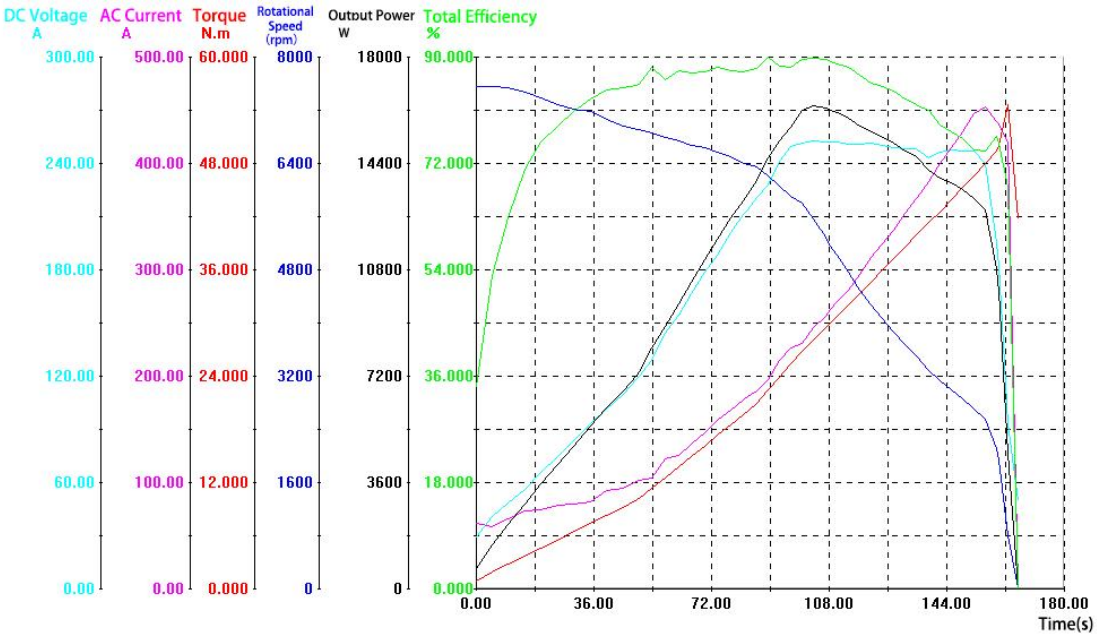






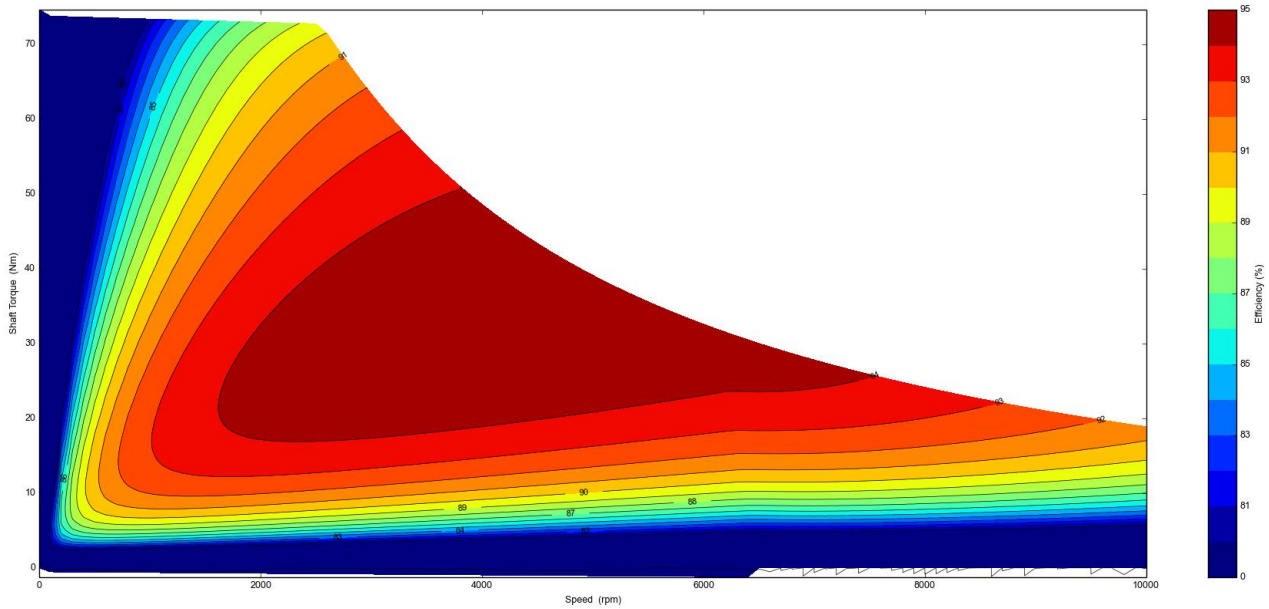
# 6.Motor test parameters：

## 165 10KW ND96850 busbar250A phase850A 7550rpm




| Characteristic point | DC voltage | DC current | DC power | AC voltage | AC current | AC power | Power factor | Frequency | Torque | Rotational speed | Output power | Controller efficiency | Motor efficiency | Overall efficiency | Time |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | V | A | W | V | A | W | | Hz | N.m | rpm | W | % | % | % | S |
| No load point | 72.563 | 29.293 | 1970.8 | 51.656 | 61.160 | 1743.4 | 0.3186 | 629.44 | 0.85 | 7550.1 | 672.86 | 88.5 | 38.6 | 34.1 | 0.0000 |
| Maximum efficiency point | 71.280 | 227.68 | 16206 | 46.132 | 195.90 | 15176 | 0.9696 | 515.57 | 22.42 | 6210.4 | 14579 | 93.6 | 96.1 | 90.0 | 89.482 |
| Maximum output power point | 72.064 | 252.59 | 18186 | 41.977 | 244.91 | 16940 | 0.9513 | 460.14 | 28.07 | 5553.6 | 16322 | 93.1 | 96.4 | 89.7 | 103.49 |
| Maximum torque point | 72.470 | 92.841 | 6726.0 | 6.4319 | 420.14 | 4331.5 | 0.9255 | 55.709 | 54.61 | 798.0 | 4562.6 | 64.4 | 105.3 | 67.8 | 162.99 |
| End point | 72.525 | 49.758 | 3605.9 | 1.3286 | 0.9951 | 0.2449 | 0.1069 | 50.539 | 41.69 | 0.1 | 0.2546 | 0.0 | 104.0 | 0.0 | 165.99 |
| Load point 1 | 72.081 | 118.33 | 8489.6 | 50.130 | 100.05 | 8026.4 | 0.9241 | 575.51 | 10.00 | 6909.3 | 7232.8 | 94.6 | 90.1 | 85.2 | 49.257 |
| Load point 2 | 71.336 | 213.80 | 15230 | 47.808 | 179.94 | 14436 | 0.9689 | 531.10 | 20.00 | 6373.2 | 13344 | 94.8 | 92.4 | 87.7 | 83.225 |
| Load point 3 | 72.054 | 251.83 | 18130 | 39.149 | 262.42 | 16885 | 0.9495 | 426.18 | 30.00 | 5151.8 | 16172 | 93.1 | 95.8 | 89.2 | 108.64 |
| Load point 4 | 72.065 | 247.43 | 17812 | 28.556 | 367.08 | 15982 | 0.8803 | 288.13 | 40.00 | 3477.9 | 14563 | 89.7 | 91.1 | 81.7 | 135.22 |
| Load point 5 | 72.538 | 182.14 | 13207 | 16.780 | 436.28 | 10824 | 0.8571 | 151.57 | 50.00 | 1932.3 | 10036 | 80.8 | 93.8 | 75.4 | 159.94 |

**MAP Diagram**




# 7.Motor performance parameters

Regarding the different usage requirements of customers, it is recommended to communicate with Company. After both parties have determined a series of important parameters such as voltage, rotational speed, motor torque, and current magnitude, obtain the motor parameter performance table that meets your company's requirements from Company.

# 8.Precautions for use

(1)It is strictly prohibited to roughly handle the motor by using excessive force, and it is also strictly prohibited to pull the lead wires of the motor.

(2)When installing the tire, you must be careful and handle it gently. Avoid scratching or causing damage to the motor hub, end cover, motor shaft, and lead wires due to excessive force.

(3)All the terminals of the phase wires and Hall elements must ensure good contact and there should be no looseness. Otherwise, it may burn out the motor and the controller. It is strictly prohibited to connect them in reverse or wrongly, as this will lead to permanent damage to the motor.

(4)It is necessary to use a controller that is matched with the motor or a controller recommended by the manufacturer to ensure the reliability of the motor's operation.

(5)It is prohibited to disassemble the motor, change its structure, or modify it. Otherwise, at the least, the warranty will become invalid, and at the worst, it may lead to unpredictable safety accidents!

(6)The installation of the motor and the flat fork must be firm and reliable. The fixing screws on the shaft part must be tightened, and there should be no looseness at all.

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